

Monitored feedback versus open-loop control for Kuramoto–XY synchronisation on IBM dynamic circuits

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Abstract

Dynamic circuits make it possible to test monitored quantum-control policies on gate-model hardware rather than only static circuit families. We report a preregistered four-qubit Kuramoto–XY feedback experiment on `ibm_kingston`. The payload uses three system qubits, one monitor qubit, three dynamic rounds, and paired hardware arms: a monitored-feedback arm and a matched open-loop control. The preregistered binary synchronisation endpoint is negative: the feedback arm gives mean target error 0.2148958333, while the matched control gives 0.2116406250, so the feedback policy does not improve the selected target. Because the binary proxy is near saturation in both arms, we add same-paper direct-observable extensions. S1b measures final XY-sector correlators on the original body; S1c reduces depth and gain; S1d and S1e sweep and repeat policy direction; and S1f tests quadrature and population controls. The extensions expose structured, policy-sensitive feedback–control differences, including a repeated positive YYI response in one shallow policy window. A later quadrature check does not reproduce that large YYI response, reducing the mean absolute feedback–control separation to 0.0082812500 and blocking a stable controller-promotion claim. The contribution is therefore a hardware-control boundary result: paired dynamic-circuit feedback tests and direct Pauli readouts can reveal calibration-window-sensitive response modes, but the tested controller is not supported as a backend-general or robust synchronisation-improving policy.

I. INTRODUCTION

Kuramoto synchronisation is a canonical model for collective phase dynamics and has a long history in nonlinear science [1, 2]. Gate-model quantum circuits can encode small XY-type phase-coupled systems, but noisy intermediate-scale quantum hardware requires conservative claim boundaries [3, 4]. In this setting, the scientific question is not whether a broad quantum advantage has been shown. The useful question is narrower: can a small monitored feedback policy improve a preregistered hardware observable relative to a matched open-loop arm under the same backend, layout, shot budget, repetitions, and calibration window?

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Measurement-conditioned feedback is a standard quantum-control primitive [5]. Dynamic circuits provide the required gate-model hardware surface: mid-circuit measurement, classical conditioning, and subsequent gates within one submitted payload [6–8]. Mid-circuit readout has also become a practical resource for near-term quantum simulation [9], and adaptive circuits can express tasks unavailable to fixed-depth unitary circuits at the same depth [10]. They also create a stricter experimental standard. A feedback arm alone is not interpretable, because backend drift, layout selection, readout bias, and ordinary shot noise can all masquerade as control response. The S1 experiment therefore uses a paired-arm design: every feedback run is compared with a matched open-loop control generated from the same circuit body and executed in the same hardware window.

This paper reports the completed S1 hardware programme. The primary preregistered binary proxy is negative. The same-paper extensions then ask whether the negative proxy hides smaller channel-level responses in direct XY-sector observables. The final result is scientifically useful, but conservative: the tested controller is not promoted. Instead, the paired-arm method identifies a structured failure and sensitivity boundary for future re-designed feedback policies.

II. PREREGISTERED S1 PROTOCOL

The primary S1 payload uses three system qubits and one monitor qubit. The feedback arm runs three dynamic rounds. In each round, the monitor qubit is coupled to the system, measured, unconditionally reset, and used to condition a small correction on the system. The matched open-loop arm preserves the same body structure without using the measured monitor outcome as an active feedback signal. The readiness artefact is generated by the committed S1 readiness script, and the canonical preregistration package is the JSON file dated May 6, 2026 under `data/s1_feedback_loop/`. The live payload used unconditional monitor reset and conditional correction only, because the submitted provider path rejected reset operations inside conditional blocks. This implementation detail is part of the experimental boundary: the circuit tests provider-supported dynamic-circuit feedback, not arbitrary low-latency real-time control.

The primary endpoint is the absolute target-order-parameter error,

$$\epsilon_{\text{target}} = |R_{\text{live}} - R_{\text{target}}|, \quad (1)$$

Protocol field	Value
System qubits	3
Monitor qubits	1
Classical bits	6
Dynamic rounds	3
Hardware arms	feedback, matched open-loop control
Primary circuits	2
Shots per circuit	1024
Repetitions per arm	12
Target synchronisation proxy	0.72

TABLE I. Primary S1 preregistered dynamic-circuit payload.

where $R_{\text{target}} = 0.72$. The feedback arm is useful under the preregistered decision rule only if it reduces this error relative to the matched open-loop control.

III. DIRECT-OBSERVABLE EXTENSIONS

The primary binary proxy saturates near one on the selected backend. To avoid over-interpreting a single nearly saturated readout, S1 includes a series of same-paper direct-observable extensions. These are not separate papers and do not change the primary endpoint. They localise the failure mode by measuring XY-sector and control observables from matched feedback/control arms.

S1b preserves the original dynamic feedback body and measures final **IXX**, **IYY**, **XXI**, and **YYI** correlators. S1c reduces the body to one dynamic round, correction angle 0.06, and base gain 0.4. S1d compares three preregistered one-round policy variants: **current_shallow_positive**, **polarity_flipped**, and **weak_positive**. S1e repeats that policy sweep at five repetitions per arm. S1f keeps the S1d/S1e-favoured shallow policy and measures **XXI**, **XYI**, **YXI**, **YYI**, and **ZZI**. The **XYI**/**YXI** rows test cross-quadrature leakage, while **ZZI** is a population-sector control.

Figure 1 gives the extension-level summary. The figure is intended to expose the same logic as the tables: S1b is mixed-sign, S1c is uniformly negative, the S1d/S1e current shallow

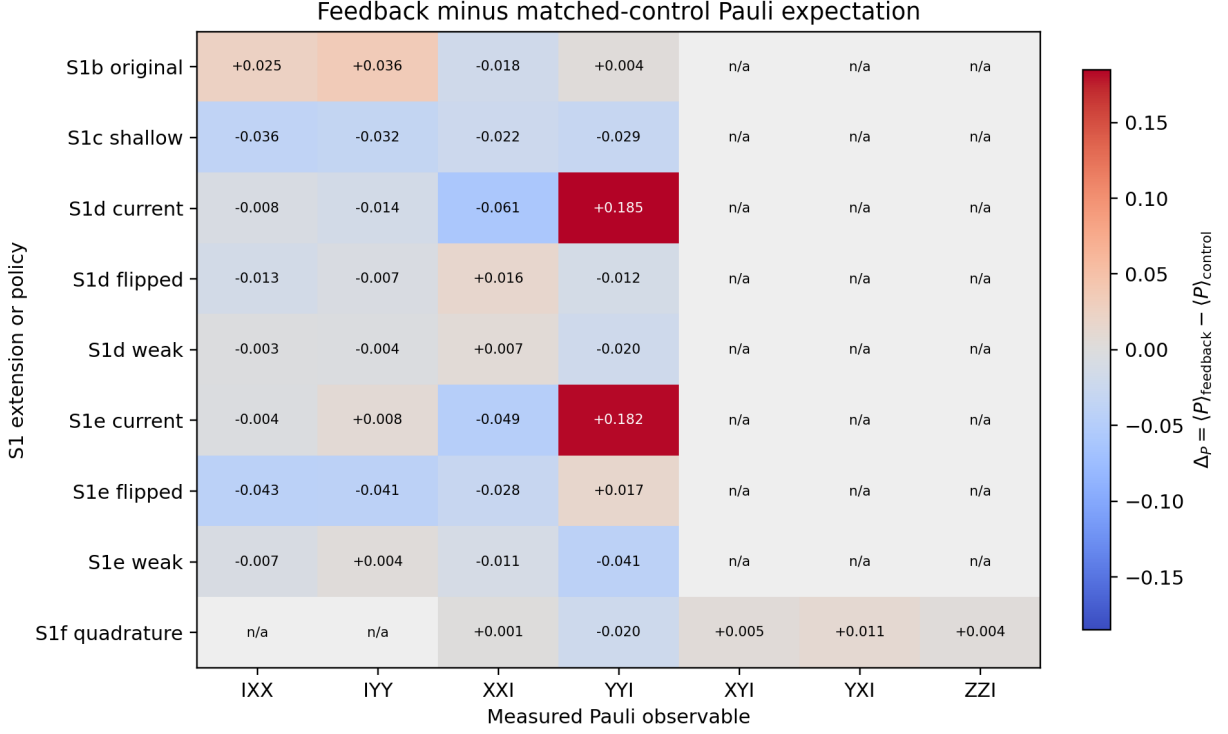


FIG. 1. Feedback-minus-control Pauli expectations across the S1 direct observable extensions. Grey cells mark observables not included in that extension. The repeated S1d/S1e YYI response and the later S1f sign reversal are visually isolated.

policy has a narrow YYI response, and S1f removes the stable positive-mechanism interpretation in the later quadrature check.

For an observable P , each arm expectation is reduced from counts as

$$\langle P \rangle = \frac{1}{N} \sum_b n_b \prod_{i \in \text{supp}(P)} (-1)^{b_i}, \quad (2)$$

with n_b the observed count for bitstring b and N the total shots. The reported extension quantity is

$$\Delta_P = \langle P \rangle_{\text{feedback}} - \langle P \rangle_{\text{control}}. \quad (3)$$

IV. PRIMARY S1 RESULT

The primary paired-arm execution completed on `ibm_kingston`. The feedback job was `d86qn3lg7okc73elg2eg`; the matched open-loop control job was `d86qn65g7okc73elg2hg`. Each arm used 12 repetitions and 12,288 total shots. The transpiled depths were 717 for

Metric	Feedback Matched control	
Mean R_{live}	0.9348958333	0.9316406250
Mean target error	0.2148958333	0.2116406250
Final R_{live}	0.9322916667	0.9322916667

TABLE II. Primary S1 binary-proxy hardware result on `ibm.kingston`.

Observable	S1b Δ_P	S1c Δ_P
IXX	0.0247395833	-0.0364583333
IYY	0.0358072917	-0.0319010417
XXI	-0.0175781250	-0.0221354167
YYI	0.0039062500	-0.0292968750
Mean $ \Delta_P $	0.0205078125	0.0299479167

TABLE III. S1b direct XY-sector extension and S1c shallow/gain-tuned extension. S1b is mixed-sign; S1c is negative in all four channels.

the feedback arm and 684 for the matched control.

The feedback arm has a slightly larger mean synchronisation proxy than the matched control, but both arms lie above the target. Consequently the feedback arm is farther from the target. The target-error improvement is -0.0032552083 , corresponding to a relative improvement of -1.5380829334% . The preregistered decision is therefore `null_or_negative`. This is the first claim boundary of the paper: the tested three-round feedback payload does not improve the selected binary-phase synchronisation proxy on this backend and calibration window.

V. XY-SECTOR LOCALISATION

S1b asks whether the negative binary endpoint hides channel-level differences in direct XY-sector observables. The original dynamic feedback body is preserved, but the final basis is rotated to the selected Pauli correlator. The results are shown in Table III.

S1b is not a positive controller result. Three channels move in the feedback direction and

Run	Variant	IXX	IYY	XXI	YYI	Mean signed	Mean abs.
S1d	<code>current_shallow_positive</code>	-0.007813	-0.013672	-0.060547	0.184896	0.025716	0.066732
S1d	<code>polarity_flipped</code>	-0.013021	-0.006510	0.015625	-0.012370	-0.004069	0.011882
S1d	<code>weak_positive</code>	-0.002604	-0.003906	0.006510	-0.020182	-0.005046	0.008301
S1e	<code>current_shallow_positive</code>	-0.003516	0.008203	-0.048828	0.182422	0.034570	0.060742
S1e	<code>polarity_flipped</code>	-0.042969	-0.040625	-0.028125	0.016797	-0.023730	0.032129
S1e	<code>weak_positive</code>	-0.007422	0.004297	-0.010547	-0.040625	-0.013574	0.015723

TABLE IV. S1d policy-direction sweep and S1e higher-repetition repeat. Values are feedback-minus-control Pauli expectations.

one moves against it, with mean absolute separation only 0.0205078125. It is, however, more informative than the saturated binary proxy: the feedback/control difference is channel-structured rather than uniformly absent.

S1c then tests whether the depth and gain of the original dynamic body are the problem. The shallower, lower-gain policy reduces the feedback depths to about 237, but all four XY-sector channels move negative. The direct-observable response therefore cannot be rescued by this shallow/gain-tuned policy.

VI. POLICY-DIRECTION SWEEP AND REPEAT

S1d compares three one-round policy variants. S1e repeats the same policy sweep at five repetitions per arm. The purpose is to distinguish a genuine policy-direction response from a single noisy calibration window. The results are shown in Table IV. The artefact timestamps place S1d at 13:46:14Z and S1e at 14:09:19Z on May 20, 2026, a separation of about 23 minutes under the same backend family and day.

The same narrow feature appears in both S1d and S1e: `current_shallow_positive` has a large positive YYI separation, 0.1848958333 in S1d and 0.1824218750 in S1e. The alternative policies remain close to zero or negative by mean signed separation. This is a real policy-sensitive signal candidate, but it is too narrow for controller promotion: IXX and XXI remain negative for the same policy, and the favourable aggregate is driven by one channel.

Observable	Feedback	Control	Δ_P
XXI	0.2187500000	0.2175781250	0.0011718750
XYI	0.1617187500	0.1566406250	0.0050781250
YXI	0.2242187500	0.2136718750	0.0105468750
YYI	-0.0445312500	-0.0242187500	-0.0203125000
ZZI	0.6523437500	0.6480468750	0.0042968750
Mean $ \Delta_P $	0.0082812500		

TABLE V. S1f quadrature and population-control check. The large positive YYI response seen in S1d/S1e does not persist in this later window.

VII. QUADRATURE MECHANISM CHECK

S1f is the final same-paper discriminator. It keeps `current_shallow_positive` and tests whether the repeated YYI response belongs to a stable quadrature structure. It measures the original XXI/YYI pair, the cross-quadrature controls XYI/YXI, and a population control ZZI. The result is shown in Table V. Its artefact timestamp is 14:22:13Z, about 13 minutes after S1e and about 103 minutes after the primary S1 paired-arm timestamp. This is therefore a later same-day calibration-window check, not a multi-day stability claim.

The S1f result blocks the strongest possible positive interpretation of S1d and S1e. The YYI separation changes sign and the mean absolute feedback–control separation falls to 0.0082812500. The cross-quadrature controls are small, and the population control is also small. The conservative interpretation is calibration-window-sensitive response, not a stable quadrature-localised feedback mechanism.

VIII. REPRODUCIBILITY ARTEFACTS

The hardware evidence is committed as raw-count and analysis artefacts under `data/s1_feedback_loop/`. Table VI lists the primary analysis artefacts and SHA256 digests used for this manuscript. The exact hardware job identifiers are retained in the JSON files and in the Markdown draft notes.

Run	Analysis artefact	SHA256
S1	s1_feedback_analysis_summa ry_ibm_kingston_20260520T1 23941Z.json	dcfae0016806ac3f705ce9ecad62f1ab47d0 d3986775170a7f15b380f38fcf24
S1b	s1b_xy_observable_analysis _ibm_kingston_20260520T130 238Z.json	91fdedc8043d25c68f95a57c21be7ed828c7 e83f34af48f11010fae686e4b152
S1c	s1c_xy_observable_analysis _ibm_kingston_20260520T132 455Z.json	ade9a407dec9759ed4fe789f738d4ff3560f 2ecf73561f6f07435c4bab9f50dd
S1d	s1d_xy_observable_analysis _ibm_kingston_20260520T134 614Z.json	6e6bc2f513f23633c72756016392b58cbf14 0d89c6f9cc7fdd07880c66b61c76
S1e	s1e_xy_observable_analysis _ibm_kingston_20260520T140 919Z.json	4f7e106e2af8e42aa74c9c07d66032bc76a7 8a936c499370c56fcef8c32ad522
S1f	s1f_xy_observable_analysis _ibm_kingston_20260520T142 213Z.json	17b37753460137c796a5fb8c52784ce694d7 ce4ffb63e55c92853abe8f91313e

TABLE VI. Committed analysis artefacts supporting the S1–S1f manuscript.

IX. CLAIM BOUNDARY

The supported claims are deliberately narrow. First, the S1 monitored-feedback payload and matched open-loop control executed on `ibm_kingston` and the feedback arm did not improve the preregistered binary synchronisation target. Second, direct XY-sector extensions show that feedback/control differences can be channel-structured and policy-sensitive even when the primary binary proxy is negative. Third, the repeated S1d/S1e YYI response is not stable under the later S1f quadrature check, so the controller is not promoted.

The paper does not claim quantum advantage, backend-general feedback control, full real-time control beyond provider-supported dynamic circuits, scalable control, or larger-

system synchronisation. It also does not claim that the observed response is isolated from calibration drift, layout-dependent coherent error, readout effects, or finite-shot variation. The result is a bounded hardware-control measurement and a falsifier for the tested feedback law.

The current S1 hardware artefacts do not include independent readout calibration circuits for S1b–S1f. For that reason, tensor-product or mid-circuit-aware readout mitigation cannot be replayed from this committed data package without adding new calibration executions. This is an explicit limitation rather than a hidden post-processing omission. Mid-circuit measurement and feedforward readout mitigation is an active technical area [11]; the correct follow-up is a preregistered S1g lane with matched feedback/control arms, calibration circuits in the same submitted block, and a fixed mitigation reducer before execution. A ZNE subset would also be useful, but it should be executed as a preregistered stress test rather than fitted after inspecting the present extensions.

X. DISCUSSION

The central scientific value of S1 is methodological rather than promotional. The primary endpoint fails, and that failure matters: it prevents the dynamic feedback policy from being sold as a working synchronisation controller. The extensions then add mechanism information that a binary endpoint alone would hide. S1b shows mixed XY-sector response on the original body; S1c shows that a shallower/lower-gain body is uniformly negative across the same channels; S1d and S1e show a repeatable single-channel response in one policy window; and S1f shows that this response does not survive a later quadrature mechanism check.

This sequence is useful for future control design because it converts a simple negative result into design constraints. Any next-generation S1 policy must preregister direct Pauli observables rather than rely only on the binary proxy. It must include a matched open-loop arm, policy-polarity controls, cross-quadrature controls, and a population control. It should also reserve a same-layout repeat in a later calibration window before promoting a controller. It should additionally treat provider constraints as first-class experimental variables: mid-circuit measurements and resets add latency, classical conditioning changes the transpiled control-flow path, reset fidelity can couple the monitor qubit back into subsequent rounds, and measurement crosstalk can affect both the monitored branch and the

final readout. Deterministic error-suppression and calibration-aware workflows can improve near-term algorithm performance [12], but S1 shows that the feedback law itself still needs a matched-arm falsification gate. Under these standards, S1 is not archive clutter: it is a claim-bound falsification and localisation paper for dynamic-circuit feedback on current IBM hardware.

XI. CONCLUSION

The S1 programme tests monitored feedback at the smallest scale where IBM dynamic circuits can implement an active monitor-and-correct loop with a matched open-loop control. The preregistered binary synchronisation endpoint is negative. Direct-observable extensions reveal structured, policy-sensitive feedback/control differences, but the final quadrature check does not support a stable positive controller mechanism. The appropriate publication claim is therefore precise: S1 establishes a reproducible paired-arm method and a hardware-control boundary for Kuramoto–XY feedback, while rejecting promotion of the tested controller under the observed `ibm_kingston` windows.

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